

KeraNova Data Collection Sheet

Version 1.0 for committee review on 1 October 2026

KeraNova Research Team | Version 1.0 | 1 October 2026

Purpose and use

This form supports KeraNova, a retrospective keratoconus study and exploratory prediction of twelve-month corrected distance visual acuity (CDVA). Complete one eye treatment record and repeat the visit form for every relevant visit. An empty entry space is intended for data collection; it is not a patient result.

Use coded patient IDs in the analytic file. The authorized institutional custodian maintains the linkage to source records separately. The index treatment is TG-PRK plus accelerated corneal cross-linking (CXL), or accelerated CXL alone, documented during the parent study period. The intended cohort is up to 60 eligible eyes with 30 per group. Verify the actual cohort before reporting its size.

Record the original measurement and its method before conversion. Use ND for not documented, NA for not applicable, UNK for explicitly unknown, NR for a source that cannot be retrieved, and AMBIG for an unresolved meaning or conflict. Zero and No are observed values. Do not fill missing CDVA from uncorrected or pinhole vision.

Record identity and source

Field	Entry	Required evidence
Coded patient ID	_____	Stable across both eyes and visits
Eye record ID and index episode ID	_____	Unique eye and treatment episode
Laterality	OD / OS / UNK	Right or left from source
Age at index treatment	_____ completed years	Date of birth and index date checked locally; no birth date in analytic export
Sex as documented	Male / Female / Other recorded / UNK	Do not infer from name
Source reference and extraction date	_____	File hash, source row or coded record and YYYY-MM-DD
Extractor and reviewer codes	_____	Coded staff identifiers
Index treatment date	YYYY-MM-DD _____	Operative record
Final eligibility status and reason	Eligible / Excluded / Pending	Record each reason and supporting source

Eligibility screening

Complete every criterion from dated source records. Do not infer progression or a contraindication from a single topographic measurement. The clinical lead must confirm the operational progression definition and procedure-specific thickness safety criterion before cohort selection. A missing criterion remains pending.

Criterion	Assessment	Evidence to record
Age at treatment at least 18 years	Yes / No / UNK	Verified age at index procedure
Progressive keratoconus documented	Yes / No / UNK	Progression definition, dates and findings: _____
Adequate corneal thickness for received procedure	Yes / No / UNK	Metric, measured thickness, criterion and clinical review: _____
No prior corneal surgery, including keratoplasty or laser vision correction	Yes / No / UNK	History and dated procedure records; apply approved parent eligibility criteria
No CXL contraindication	Yes / No / UNK	Clinical assessment and relevant history
No other ocular disease affecting the outcome	Yes / No / UNK	Source diagnosis and clinical adjudication
No relevant uncontrolled systemic disease	Yes / No / UNK	Parent criteria include diabetes with HbA1c >8.0% or recent ketoacidosis, and active autoimmune connective-tissue disease associated with corneal melting
Pregnancy or lactation exclusion satisfied when required	Yes / No / NA / UNK	Apply the actual IRB determination to historical treatment records; clarify inconsistent parent wording
Index treatment within approved study period	Yes / No / UNK	Parent protocol January 2020 to January 2025; verify exact approved boundary dates
Procedure matches a study cohort	Yes / No / UNK	TG-PRK plus accelerated CXL or accelerated CXL alone
Baseline and twelve-month outcome available	Yes / No / UNK	Record separately from core clinical eligibility; missing label excludes prediction analysis

Baseline clinical measurements

Use the nearest valid preoperative measurement for the index eye and preserve each measurement date. A same-day value is baseline only if the source establishes that it was measured before treatment. Record how far each measure precedes the operation and clinically review distant baseline values. Any sensitivity analysis restricting this interval must be prespecified before analysis.

Measurement	Entry	Method and quality
Visit date and time if available	YYYY-MM-DD _____	Preoperative chronology verified Yes / No / UNK
CDVA original notation	_____	Chart type, testing distance and correction method: _____
CDVA standardized value	____ logMAR / missing code	Conversion method and reviewer: _____
UDVA original and logMAR	_____	Uncorrected distance vision kept separate
Manifest sphere and cylinder	____ D and ____ D	Minus or plus cylinder convention: _____
Cylinder axis and MRSE	____ degrees; ____ D	MRSE original and calculated separately
K1 and K2	____ D; ____ D	Dated device report and scan quality

Measurement	Entry	Method and quality
Kmax	____ D	Primary candidate predictor
Thinnest corneal thickness TCT	____ micrometres	Prespecified pachymetry candidate; distinguish from central or apical thickness
BAD-D and Amsler Krumeich stage	_____	Only verified device score and documented stage
Prior surgery, scar and other ocular disease	_____	Presence, laterality and dated clinical description

Index treatment details

Field	Entry	Source requirement
Procedure as written in record	_____	Keep exact source phrase
Topography guided PRK verified	Yes / No / UNK	Operative or laser treatment report
Accelerated CXL verified	Yes / No / UNK	Operative report and irradiation protocol
Final study group	TG-PRK plus accelerated CXL / accelerated CXL alone / Other / Pending	Clinical reviewer code: _____
CXL protocol	_____	Irradiance, duration, fluence, epithelial status and riboflavin if recorded
PRK protocol	_____	Ablation depth, optical zone and relevant adjuncts if recorded
Additional procedure or deviation	_____	PTK and other procedures cannot silently enter either primary group

Repeatable follow up visit form

Capture all relevant visits with actual dates. One- and six-month source labels are interim descriptive information until dates are verified. The primary endpoint is the valid CDVA visit closest to the twelve-month anniversary between the tenth- and fourteenth-month anniversaries, inclusive. Break equal date-distance ties by choosing the earlier visit. Never select a visit because it has better vision.

Field	Entry	Required context
Visit ID and actual date	_____	YYYY-MM-DD and source reference
Source interval label and elapsed days	_____	Retain label separately from date-based selection
CDVA original and standardized	_____	Chart, correction, modifiers and logMAR
UDVA original and standardized	_____	Separate measurement and method
Sphere, cylinder, axis, MRSE	_____	D, D, degrees, D; same encounter

Field	Entry	Required context
K1, K2, Kmax and TCT	_____	D, D, D, micrometres; dated scan
Haze and other complications	Yes / No / UNK; description _____	Grade scale, presence, location, onset and interventions
Repeat procedure or new ocular event	_____	Type, date and relation to index eye
Endpoint selected	Yes / No / Pending	Reason, anniversary distance, quality and missing code

Optional imaging and extended Pentacam data

Images are an optional linked research resource. Record a coded image ID, patient ID, laterality, scan date, preoperative or postoperative status, modality, device and software version, map type, scale, units, scan-quality flag and permitted use. Remove identifiers in metadata and burned-in pixels before an authorized export. Keep original scan metadata locally and preserve a transformation log.

The supplied CSV contains no images or imaging references. Public images cannot supply this cohort's missing postoperative CDVA. Image-based experiments require a separate objective, compatible labels and patient-level splitting; they must not be presented as validation of the twelve-month prediction model.

The header-only research workbook proposes extended fields such as corneal indices, aberrometry and densitometry. Collect these when available and verified, but they are not additional predictors in the primary prespecified model.

Review and release

- Check every treatment assignment, clinical eligibility decision and primary CDVA label against the source with a second reviewer.
 - Record each unresolved issue, requested source, reviewer decision and date. Freeze cohort and dictionary versions before statistical modeling.
 - Reconcile screened eye records, unique patients, exclusions, pending records and analysis denominators.
- Signatures: extractor _____ clinical reviewer _____ data custodian _____ date _____.